

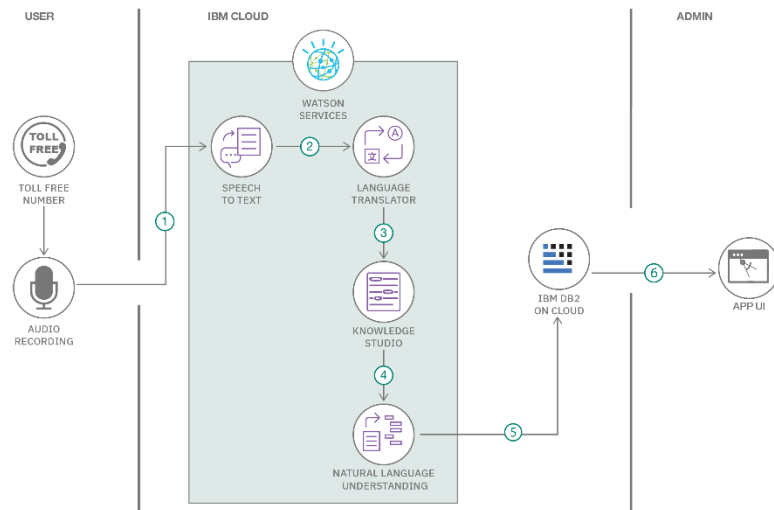
Project Design Phase-II Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table

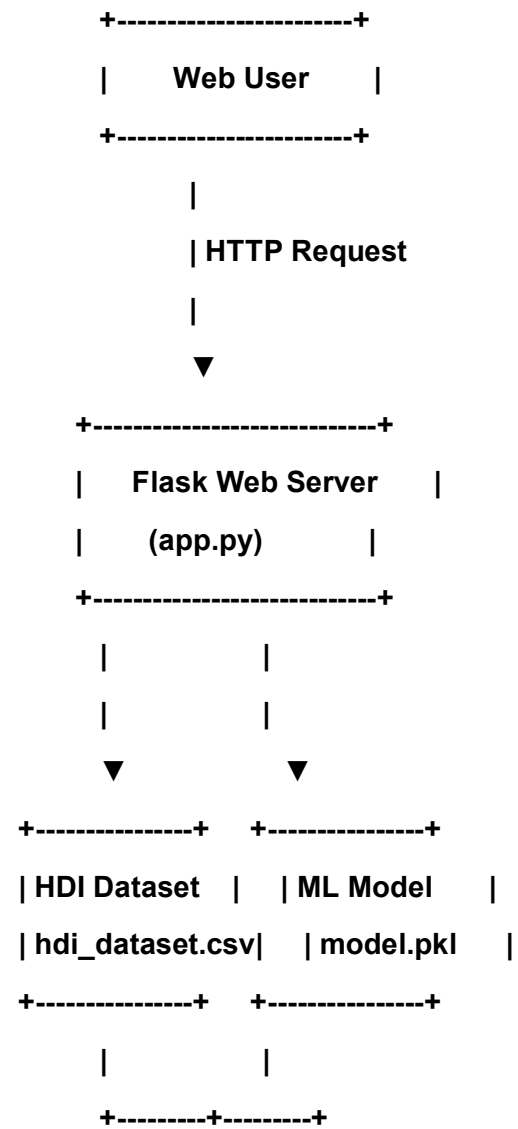
Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

DIAGRAM



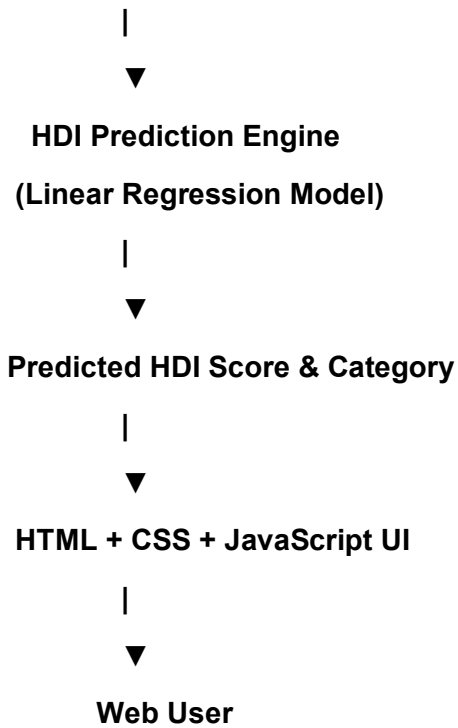


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for country selection and HDI prediction	HTML5, CSS3, JavaScript
2	Application Logic-1	Handle user requests and routing	Python, Flask
3	Application Logic-2	Read dataset and process country information	Pandas
4	Application Logic-3	Generate HDI prediction	Scikit-learn (Linear Regression)
5	Database	Country HDI data stored locally	CSV File (hdi_dataset.csv)

S.No	Component	Description	Technology
6	Cloud Database	Not Applicable	N/A
7	File Storage	Stores dataset, trained model and graphs	Local File System
8	External API-1	Not Used	N/A
9	External API-2	Not Used	N/A
10	Machine Learning Model	Predict Human Development Index	Linear Regression (Scikit-learn)
11	Infrastructure (Server / Cloud)	Local deployment using Flask development server	Local Server (Python Flask)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used to build the application	Flask, Pandas, NumPy, Matplotlib, Scikit-learn
2	Security Implementations	Input validation and safe handling of user requests	Flask Validation, Python Exception Handling
3	Scalable Architecture	Modular architecture allowing easy dataset and model updates	Flask (3-Tier Architecture)
4	Availability	Available whenever the Flask server is running	Python Flask Local Server
5	Performance	Fast prediction using a pre-trained ML model without retraining	Scikit-learn, Joblib

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>